

7.5 Strategy for Integration

1. Simplify the integrand.

Example 1

$$\int (\sin(x) + \cos(x))^2 dx$$

Example 2

$$\int \frac{2x + 1}{x^2 + 3} dx$$

2. Look for an obvious u-substitution.

Example 3

$$\int \frac{x^2}{\sqrt{x^3 - 4}} dx$$

Example 4

$$\int \tan(x) \sec^4(x) dx$$

3. Identify a technique to use based on the integrand's form.

(a) trig integrals

(b) rational functions

(c) products of algebraic and transcendental functions

(d) radicals

4. If the previous steps didn't work, keep trying!

Example 5

Integrate the following:

1. $\int \cos(\sqrt{x}) dx$

2. $\int e^{2x+e^x} dx$

Example 6

Integrate the following:

1. $\int \frac{2 \tan^3(x)}{\cos(x) + \cos(x) \cos(2x)} dx$

2. $\int \frac{dx}{x + x\sqrt{x}}$

3. $\int \frac{dx}{\sqrt{1-x^2} + 1-x^2}$